

Yukang Xie

PhD Researcher in Computer Architecture · Functional Languages · FPGA

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Research Experience

Architectural Optimisation for Lazy Functional Language Processors

2025–2026

- Designed a **pipelined, dataflow-driven processor** supporting automatic multithreading.
- Designed a companion **hardware garbage collector** with concurrent mark-and-sweep.
- Achieved up to **118% runtime improvement** over the single-cycle design, with zero measured GC overhead on general benchmarks.

Specialised Processor for Haskell Based on Structured Combinators

2024–2025

- Designed a single-cycle stack machine for efficient combinator graph reduction, executing one structured combinator per cycle.
- Extended the **MicroHs** compiler with a new code-generation scheme for structured combinators.
- Achieved a minimal FPGA area footprint while reducing runtime by up to **58%** relative to SKI combinator schemes.

Reconfigurable Architecture for Cryptography Processing (Master's Thesis)

2023–2024

- Integrated heterogeneous coarse-grained cryptographic operation units through on-chip networks.
- Designed a programming model and DSL for expressing cryptographic protocols and algorithms in a dataflow manner.
- Mapped DSL descriptions onto the architecture and tuned the on-chip network through simulation and FPGA prototyping.

Virtualisation Support for Symmetric-Cipher Hardware Accelerators

2022–2023

- Designed schemes that partition AES/SM4 cipher pipelines into independent units for parallel execution of tasks from different guests.
- Supported guest-level parallelisation while retaining throughput/area efficiency comparable to prior work.

Selected Publications

- ASAP 2026.** *Ouros: A Dataflow-Driven Processor for Lazy Functional Programming Languages.*
- TFP 2025.** *From Haskell to a New Structured Combinator Processor.*
- ICA3PP 2023.** *A Pipelined AES and SM4 Hardware Implementation for Multi-Tasking Environments.*

Education

Heriot-Watt University, Edinburgh

Sep 2024–Present

PhD in Computer Science

Wuhan University, Wuhan

Sep 2021–Jun 2024

MEng in Cyberspace Security

GPA: 87.52/100

Huazhong University of Science and Technology, Wuhan

Sep 2017–Jun 2021

BEng in Hydraulic and Hydro-Power Engineering

GPA: 3.41/4.00

Technical Skills

Programming: Haskell, Rust, Scala/Chisel HDL (happier with functional programming, but comfortable with other languages)

Hardware : Simulator implementation, FPGA prototyping and verification, Xilinx FPGA toolchains

Architecture : Functional-language processors, cryptographic accelerators, on-chip networks, CGRAs, conventional CPU microarchitecture

Verification : Hardware correctness model checking with Kind 2

Languages : Chinese (native), English (IELTS 7.5)